# Probability

- Probability

→ Probability is a mathematical tool which measures the possibilities of happenning or non-happenning of an event.

- Some basic terminologies:

1. Random experiment:
2. Event or outcome:
3. Sample space:
4. Exhaustive cases:
5. Mutually Exclusive Events:
6. Equally Likely events:
7. Favourable cases:
8. dependent and Independent events

- Mathematical Definition of probability.

→ If $n$ be the mutually Exclusive equally likely exhaustive number of cases. of which $m$ cases are favourable to the occurrence of an event E, then the probability of occurrence of that event E is denoted by P(E) and is given by

$$P(E) = \frac{\text{favourable no. of cases}}{\text{Total no. of cases}} = \frac{m}{n}$$

> Properties:

1. If $P(\bar{E})$ denotes the probability of non-occurrence of an event E, then $P(E) + P(\bar{E}) = 1$

2. P(sure / certain event) = 1
P(Impossible event) = 0

3. $0 \leq P(E) \leq 1$

# Basic Theorems on Probability:

1. Addition Theorem (Law of total probability)

Let, A and B be any two events, then the probability of occurrence of at-least one of the event A or B is denoted by $P(A \cup B)$ and is given by
$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Where,
$P(A \cap B)$ denotes the simultaneous occurrence of both the events A and B.

2. Multiplication Theorem (Law of compound ~~pro~~ probability).

↳ let A and B be two independent events, the probability of simultaneous occurrence of both of the events A and B is denoted by $P(A \cap B)$ and is given by

$$P(A \cap B) = P(A) \times P(B)$$

Notes:

① If A and B be two mutually exclusive events, then

- $P(A \cap B) = 0$
- $P(A \cup B) = P(A) + P(B)$

② For independent events

- $P(A \cup B) = 1 - P(\bar{A}) \times P(\bar{B})$
- $P(A \cup B \cup C \cup D) = 1 - P(\bar{A}) \times P(\bar{B}) \times P(\bar{C}) \times P(\bar{D})$

~~Exer~~

Exercise - 16

② A card is drawn from a well-shuffled deck of 52 cards. What is the probability that it is,

↳ Soln,

Total no. of cards $n(S) = 52$

a) No. of heart $n(H) = 13$

$$\therefore P(H) = \frac{n(H)}{n(S)} = \frac{13}{52} = \frac{1}{4}$$

b) No. of Jacks $n(J) = 4$

No. of spade $n(Sp) = 13$

$$\cancel{\therefore P(J) = \frac{n(J)}{n(S)} = \frac{4}{52}}$$

No. of common card $n(J \cap S_p) = 1$

$$\therefore P(J \text{ or } S_p) = P(J) + P(S_p) - P(J \cap S_p)$$

$$= \frac{n(J)}{n(S)} + \frac{n(S_p)}{n(S)} - \frac{n(J \cap S_p)}{n(S)}$$

$$= \frac{4}{52} + \frac{13}{52} - \frac{1}{52}$$

$$= \frac{16}{52}$$

$$= \frac{4}{13}$$

~~Three~~

c) a black card

no. of black card $n(B) = 26$

$$\therefore P(B) = \frac{n(B)}{n(S)} = \frac{26}{52} = \frac{1}{2}$$

d) a face card

no. of face card $n(F) = 12$

$$\therefore P(F) = \frac{n(F)}{n(S)} = \frac{12}{52} = \frac{3}{13}$$

5. Three fair coins are tossed at the same time. Find the probability of getting.

→ Soln,

- The sample space for tossing 3 coins is
  S = {HHH, THH, HTH, HHT, HTT, THT, TTH, TTT}

∴ Total number of samples n(s) = 8

(a) All tails

→ Let A be the event of getting all tails.

A = {TTT}

So, n(A) = 1

$$P(A) = \frac{n(A)}{n(S)} = \frac{1}{8}$$

(b) One head and two tails

→ B = {HTT, THT, TTH}

n(B) = 3

~~So,~~ ∴ $P(B) = \frac{n(B)}{n(S)} = \frac{3}{8}$

(c) At least one tail

→ Let C be the event of getting at least one tail.

C = {HHH, THH, HTH, HHT, HTT, THT, TTH}

So, n(s) = 7

$$P(C) = \frac{n(C)}{n(S)} = \frac{7}{8}$$

∴ $P(C) = \frac{7}{8}$

c) at most one tail

→ let D be the event of getting at most one tail.

$D = \{HHH, THH, HTH, HHT\}$

so, $n(D) = 4$

Now,

$$P(D) = \frac{n(D)}{n(S)} = \frac{4}{8} = \frac{1}{2}$$

7) Following is the mark distribution of 60 students of a examination. If marks is selected at random, find the probability of getting a student who obtain the marks.

→

| Marks | 0-10 | 10-20 | 20-30 | 30-40 | 40-50 |
|---|---|---|---|---|---|
| No. of Students | 10 | 15 | 12 | 12 | 11 |

Total no. of students $(n) = 10 + 15 + 12 + 12 + 11 = 60$

a) ~~Ea~~ less than 20

→ Favourable no. of students for less than 20 marks $(m_1) = 10 + 15 = 25$

$$\therefore P(\text{less than } 20) = \frac{m_1}{n} = \frac{25}{60} = \frac{5}{12}$$

b) more than or equal to 30

↳ Favourable no. of students for more than or equal to 30 marks

$(m_2) = 12 + 11 = 23$

$\therefore$ P(more than or equal to 30) $= \frac{m_2}{n}$

$= \frac{23}{60}$ #.

c) more than or equal 20 and less than 40.

↳ No. of students who obtained more than or equal to 20 and less than 40 marks

$(m_3) = 12 + 12 = 24$

$\therefore$ P(more than or equal 20 and less than 40) $= \frac{m_3}{n} = \frac{24}{60} = \frac{2}{5}$ #

8. Two dice are thrown together. Find the probability of getting.

↳ Soln,

Total no. of cases for throwing 2 dice

$n(S) = 36$

a) both even digits.

↳ let A be the event of getting both even digits.

$A = \{(2,2), (2,4), (2,6), (4,2), (4,4), (4,6), (6,2), (6,4), (6,6)\}$

So, $n(A) = 9$

2 2
3 3
4 4
5 5
6 6



$\therefore P(A) = \frac{n(A)}{n(S)} = \frac{9}{36} = \frac{1}{4}$ #.

b) a total of 8.
→ let B be the event of getting a total of 8.
$B = \{(2,6), (4,4), (3,5), (5,3), (6,2)\}$
so, $n(B) = 5$

$\therefore P(B) = \frac{n(B)}{n(S)} = \frac{5}{36}$ #.

10.
a) A leap year is selected at random. What is the probability that it will contain 53 sundays?
→ A leap year will contain 366 days.
i.e. 366 days $= (52 \times 7 + 2)$ days
$= 52$ weeks $+ 2$ days

so,
for a leap year to contain 53 sundays, these extra two days must contain a sunday. The possibilities for these extra two days are as follows:

(Sun, Mon), (Mon, Tue), (Tue, Wed), (Wed, Thu), (Thu, Fri), (Fri, Sat), (Sat, Sun)
Here,
Out of 7 possibilities only 2 will contain sunday.
$\therefore P(53 \text{ sundays}) = \frac{\text{Favourable no. of cases}}{\text{Total no. of cases}}$

$= \frac{2}{7}$ #.

b) What is the probability that there will be 5 saturdays in the month of May selected at random?

→ Soln,

There are 31 days in month of May

$30 = 7 \times 4 + 3$ days

$= 4$ week $+ 3$ days.

So,

for the month of may ~~will~~ have 5 saturdays, The possibilities for these two extra days are as follow:

(Sat, sun, mon), (sun, mon, tue), (mon, tue, wed)
(tue, wed, thu), (wed, thu, Fri), (thu, Fri, sat)
(Fri, sat, sun)

Here,

Out of 7 possibilities only 3 will contain saturday.

$\therefore P(5 \text{ saturday}) = \frac{\text{Favourable no. of cases}}{\text{Total possible outcomes}}$

$= \frac{3}{7}$ #.

9.

b) A problem in mathematics is given to three students X, Y, Z, whose chances of solving it are $\frac{3}{4}$, $\frac{4}{5}$, $\frac{5}{6}$ respectively.

Find the probability that:

i) None of them will solved the problem.

ii) the problem will be solved.

iii) The Problem will be solved by exactly one of them.

→ Soln,

$$P(X) = \frac{3}{4} \Rightarrow P(\bar{X}) = 1 - \frac{3}{4} = \frac{1}{4}$$

$$P(Y) = \frac{4}{5} \Rightarrow P(\bar{Y}) = 1 - \frac{4}{5} = \frac{1}{5}$$

$$P(Z) = \frac{5}{6} \Rightarrow P(\bar{Z}) = 1 - \frac{5}{6} = \frac{1}{6}$$

i) P(none will solve the problem) = $P(\bar{X} \cap \bar{Y} \cap \bar{Z})$

$$= P(\bar{X}) \times P(\bar{Y}) \times P(\bar{Z})$$

$$= \frac{1}{4} \times \frac{1}{5} \times \frac{1}{6}$$

$$= \frac{1}{120}$$ Ans.

ii) P(problem will be solved) = $1 - P(\bar{X} \cap \bar{Y} \cap \bar{Z})$

$$= 1 - \frac{1}{120}$$

$$= \frac{119}{120}$$ Ans.

iii) P(exactly one will solve) = $P(X \cap \overline{Y} \cap \overline{Z}) + P(\overline{X} \cap Y \cap \overline{Z}) + P(\overline{X} \cap \overline{Y} \cap Z)$

$= P(X) \cdot P(\overline{Y}) \cdot P(\overline{Z}) + P(\overline{X}) \cdot P(Y) + P(\overline{Z}) + P(\overline{X}) \cdot P(\overline{Y}) \cdot P(Z)$

$= \frac{\cancel{3}}{4} \times \frac{1}{5} \times \frac{1}{\cancel{6}_2} + \frac{1}{\cancel{4}} \times \frac{\cancel{4}}{5} \times \frac{1}{6} + \frac{1}{4} \times \frac{1}{\cancel{5}} \times \frac{5}{6}$

$= \frac{1}{40} + \frac{1}{30} + \frac{1}{24}$

$= \frac{1}{10}$ #

11.

c) A class consists of 20 boys and 30 girls. If three students are choosen at random, what is the probability that
i) all are boys.
ii) all are girls.
iii) all students of same gender.
iv) only first is boy.

→ Soln,

No. of boys $n(B) = 20$
No. of girls $n(G) = 30$
No. of students $n(S) = 50$

Here, 3 students are choosen at random.

i) P(All are Boys) $= P(B_1 \cap B_2 \cap B_3)$

$= \cancel{\frac{20}{50}} \cdot P(B_1) \times P(B_2) \times P(B_3)$

$= \frac{20}{50} \times \frac{19}{49} \times \frac{18}{48}$

$= \frac{57}{980}$

ii) $P(\text{All are girls}) = P(G_1 \cap G_2 \cap G_3)$

$= P(G_1) \times P(G_2) \times P(G_3)$

$= \frac{30}{50} \times \frac{29}{49} \times \frac{28}{48}$

$= \frac{29}{140}$

iii) $P(\text{All students of same gender}) =$

$P(B_1 \cap B_2 \cap B_3) + P(G_1 \cap G_2 \cap G_3)$

$= \frac{57}{980} + \frac{29}{140}$

$= \frac{13}{49}$ #.

iv) $P(\text{only first is boy}) = P(B_1 \cap G_2 \cap G_3)$

$= P(B_1) \times P(G_2) \times P(G_3)$

$= \frac{20}{50} \times \frac{30}{49} \times \frac{29}{48}$

$= \frac{29}{196}$ #.

1) A die is rolled once. Find the probability of getting.

a) 2 or 5 in turning up

↳ Soln,

$P(2 \text{ or } 5) = P(2) + P(5)$

$= \frac{1}{6} + \frac{1}{6}$

$= \frac{2}{6}$

$= \frac{1}{3}$ #

b) a number greater or equal to 4.

→ Soln,

P(Number greater or equal to 4) = $P(4) + P(5) + P(6)$

$= \frac{1}{6} + \frac{1}{6} + \frac{1}{6}$

$= \frac{3}{6}$

$= \frac{1}{2}$

c) an odd numbers.

→ P(Odd numbers) = $P(1) + P(3) + P(5)$

$= \frac{1}{6} + \frac{1}{6} + \frac{1}{6}$

$= \frac{3}{6}$

$= \frac{1}{2}$

3) A ball is drawn at random from a bag containing 12 red, 10 green and 7 black balls. What is the probability of getting.

→ Soln,

$n(R) = 12$

$n(G) = 10$

$n(B) = 7$

$n(S) = 29$

a) a green ball

$P(G) = \frac{n(G)}{n(S)} = \frac{10}{29}$

b) black ball

→ $P(B) = \frac{n(B)}{n(S)} = \frac{7}{29}$

c) not a red ball

→ $P(\bar{R}) = ?$

First,

$P(R) = \frac{n(R)}{n(S)} = \frac{12}{29}$

Now,

$P(\bar{R}) = 1 - \frac{12}{29} = \frac{17}{29}$

∴ P(not a red ball) $= \frac{17}{29}$

d) a red or black ball

→ $P(B \text{ or } R) = P(B) + P(R)$

$= \frac{7}{29} + \frac{12}{29}$

$= \frac{19}{29}$

4) From 30 tickets marked with the first 30 ~~number~~ numerals, one is drawn at random. Find the probability that

a) it is a multiple of 5 or 7.

b) a cube number.

→ Soln,

$n(S) = 30$

a)

→ let 5 be the event event of getting

Multiple of 5 and 7 be the event of getting multiple of 7.

$S = \{5, 10, 15, 20, 25, 30\} \Rightarrow n(5) = 6$

$7 = \{7, 14, 21, 28\} \Rightarrow n(7) = 4$

Now,

$$p(5 \text{ or } 7) = \frac{n(5)}{n(S)} + \frac{n(7)}{n(S)}$$

$$= \frac{6}{30} + \frac{4}{30}$$

$$= \frac{1}{5} + \frac{2}{15}$$

$$= \frac{5}{15} = \frac{1}{3}$$

b) Let C be the event of getting cube number.

$C = \{1, 8, 27\} \Rightarrow n(C) = 3$

Now,

$$P(C) = \frac{n(C)}{n(S)} = \frac{3}{30} = \frac{1}{10}$$

6)

a) If $P(A) = 0.5$, $P(B) = 0.2$ and A and B are independent events, find the probability of occurrence of at least one of the events A or B.

↳ Soln,

$P(A) = 0.5$

$P(B) = 0.2$

For independent events,

$P(A \cap B) = P(A) \times P(B)$
$= 0.5 \times 0.2$
$= 0.1$

Now,
P(At least one of A or B) $= P(A \cup B)$
$= P(A) + P(B) - P(A \cap B)$
$= 0.5 + 0.2 - 0.1$
$= 0.6$

b) If $P(A) = 0.4$, $P(B) = 0.6$ and $P(A \cup B) = 0.8$, then find $P(A \cap B)$. Are the events A and B independent?

↳ Soln,
$P(A) = 0.4$
$P(B) = 0.6$
$P(A \cup B) = 0.8$

We know that,
$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
$0.8 = 0.4 + 0.6 - P(A \cap B)$
or, $P(A \cap B) = 1 - 0.8$
$\therefore P(A \cap B) = 0.2$

Now,

$P(A) \times P(B)$
$= 0.4 \times 0.6$
$= 0.24$

Since the value of $P(A \cap B) \neq$ the value of $P(A) \times P(B)$ ~~the~~.
So the given events A and B are dependent.

9.a) A problem is given to two students A and B whose chances of solving are $\frac{1}{3}$ and $\frac{1}{4}$ respectively. What is the probability that the problem will be solved?

→ Soln,

$P(A) = \frac{1}{3}$

$P(B) = \frac{1}{4}$

$P(A) \times P(B) = \frac{1}{3} \times \frac{1}{4} = \frac{1}{12}$

Now,

$P(\text{Problem will be Solved}) = P(A) + P(B) - P(A) \times P(B)$

$= \frac{1}{3} + \frac{1}{4} - \frac{1}{12}$

$= \frac{7}{12} - \frac{1}{12}$

$= \frac{6}{12}$

$= \frac{1}{2}$

11.

a) From a pack of 52 cards, two are drawn one by one without replacement. Find the probability that.

→ Soln

$n(S) = 52$

i) both of them are jacks

→ $P(J \cap J) = P(J) \times P(J)$

$$= \frac{4}{52} \times \frac{3}{51}$$

$$= \frac{1}{221}$$ //.

ii) First King and second Jack

$P(KJ) = P(K) \times P(J)$

$$= \frac{4}{52} \times \frac{4}{51}$$

$$= \frac{4}{663}$$ //.

iii) One king and one Jack

$P(KJ \cup JK) = P(K) \times P(J) + P(J) \times P(K)$

$$= \frac{4}{663} + \frac{4}{663}$$

$$= \frac{8}{663}$$ //.

b) A bag contain 10 red and 6 white balls. Two balls are drawn at random one after another with replacement. Find the probability that

↳ Soln,

$n(R) = 10$

$n(W) = 6$

$n(S) = 16$

i) both are red ball

↳ $P(R \cap R) = P(R) \times P(R)$

$$= \frac{10}{16} \times \frac{10}{16}$$

$$= \frac{25}{64} \text{ Ans.}$$

ii) balls of different colours.

→ $P(\text{different colours}) = P(W) \times P(R) + P(R) \times P(W)$

$$= \frac{6}{16} \times \frac{10}{16} + \frac{10}{16} \times \frac{6}{16}$$

$$= \frac{15}{64} + \frac{15}{64}$$

$$= \frac{15}{32}$$

iii) balls of same colour.

→ $P(\text{Balls of same color}) = 1 - P(\text{different colours})$

$$= 1 - \frac{15}{32}$$

$$= \frac{17}{32} \text{ Ans.}$$